Patrick Moehrke

Actuarial Modelling & Machine Learning

Associate Actuary (AMASSA)

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Location: Cape Town, South Africa

I develop models in Julia and Python and build the software needed to put them into use. As an Associate Actuary at Momentum Corporate, I bring insurance and investment knowledge, with research experience in deep learning and Bayesian methods.

TECHNICAL SKILLS

Languages
Julia, Python, SQL, R, LaTeX; Zig (*basic*)

ML & Scientific Computing
Deep Learning, Probabilistic Programming, Automatic Differentiation, SciML, High Performance Computing

Modelling Libraries
scikit-learn, XGBoost, SHAP, Lux.jl, Turing.jl, ModelingToolkit.jl

Performance
Benchmarking, multithreading, allocation profiling

Systems
Linux, Bash scripting, Containers, QEMU/KVM, Kubernetes (*basic*)

Engineering & CI/CD
GitLab CI/CD, GitHub Actions, Azure DevOps; unit testing

Cloud
Terraform, architecture diagrams; AWS Lambda, ECS/Fargate, EC2, S3

AI & Workflow
Copilot Studio, Power Query, Power Automate, Jira, SharePoint

QUALIFICATIONS

Associate Actuary (AMASSA)
Actuarial Society of South Africa
13/15 actuarial exams passed
Awaiting F204 results

AWS Certified Solutions Architect
Associate

AWS Certified Cloud Practitioner

EXPERIENCE

Product Specialist (Smoothed Bonus Funds)

Momentum Corporate
May 2024 – Present | Cape Town
Structured Investments and Annuities

Model development & production delivery

- Rebuilt Excel/VBA models in Julia and Python with Git version control, documentation, automated tests and profiling.
- Developed Julia and Python financial model packages, with GitLab CI/CD for linting, tests and container image builds.
- Improved the monthly bonus BAU process by moving Excel workflows to Python and PostgreSQL, with SharePoint for team access. Built Power Query dashboards in Excel for business monitoring.
- Delivered cloud backtesting through SharePoint, with Power Automate triggering model runs from user inputs.
- Reviewed and developed smoothed bonus products for FundsAtWork and stand-alone corporate clients.
- Built a multi-asset ESG for daily backtesting and a CDS spread model using Bayesian jump stochastic differential equations.

Infrastructure & performance

- Managed AWS infrastructure with Terraform and documented architecture in diagrams. Deployed Lambda and ECS/Fargate workloads and sized resources to control costs.
- Benchmarked and profiled Julia and Python code, analysing memory allocations before deployment.

Governance & IT collaboration

- Reviewed and tested colleagues' models and code. Prepared reports for internal stakeholders and presented findings to Client ALM and Momentum Corporate PMC.
- Defined requirements in Jira and collaborated with IT stakeholders on Orbit system changes and user acceptance testing for smoothed bonus products.

AI & team development

- Developed bots in Microsoft Copilot Studio and participated in internal AI forums. Introduced GitLab, AWS and CI/CD to the team and trained colleagues in version control, testing and deployment.

EDUCATION

MCom Actuarial Science
Stellenbosch University
2024–2026
Thesis: Bayesian neural networks for mortality modelling.

BCom (Hons) Actuarial Science
Stellenbosch University
2020
Research: genetics and risk premiums, applied to Huntington's disease.

BCom Actuarial Science
Stellenbosch University
2017–2019

AWARDS

Khuthaza Quarterly Award (4×)
Momentum Corporate

Golden Key International Honour Society
Stellenbosch University

Dean's Merit List
Six subjects, Stellenbosch University

GITHUB

github.com/patrickm663

BNN-sparse-mortality / RNNs/LSTMs
Hybrid mortality models in Julia

LocalGLMnet.jl
Neural network interpretability

zigsal / hello-sisal
SISAL revival with Zig; K&R C compatibility

HMD.jl
Mortality data package, tests and documentation

sisal-benchmarks
Julia/Python/SISAL benchmarks

moehrke-co
Portfolio site and AWS deployment

EXPERIENCE (CONTINUED)

Actuarial Consultant

Dupro Advisory (Actuartech)
July 2020 – April 2024 | Stellenbosch

Promoted to Actuarial Consultant in December 2021 after trainee and junior roles.

Deployment & reporting

- Designed and deployed serverless IFRS 17 reporting on AWS Lambda at lower cost than off-the-shelf products.
- Built a serverless web application that handled over 3 000 API calls in three days with zero failures.
- Developed and tested IFRS 17 calculations, helped deliver retrospective opening balances under GMM and VFA, and drafted methodology with external auditors.

Machine learning & explainability

- Built machine learning proofs of concept using AWS SageMaker and scikit-learn, and researched Bayesian and deep learning methods in Julia for mortality modelling.
- Researched SHAP explainability, fairness and proxy discrimination using driver telematics data.

Architecture & technical training

- Reviewed a multinational reinsurer's actuarial reporting architecture, documented risks and proposed improvements. Monitored cloud costs and investigated Kubernetes cost reduction on AWS.
- Trained South African and UK actuarial teams in Python, SQL, GitHub Actions, Azure DevOps and serverless AWS. Co-presented at ASSA and IFoA conferences.

RESEARCH & PERSONAL PROJECTS

- **Hybrid mortality models:** combined Lee–Carter, Gompertz and Heligman–Pollard models with small neural networks; trained Bayesian variants and compared HMC with ADVI.
- **Neural network explainability:** explored gradients, feature effects and symbolic regression using Lux.jl and CUDA.jl.
- **SISAL revival:** revived the legacy SISAL toolchain on modern Linux using Zig's C compiler, resolving K&R C compatibility issues and tuning multithreaded execution for current CPUs.
- **Personal GPU benchmarks:** benchmarked Lux.jl mortality models using CUDA.jl, recording training time, GPU memory use and validation error.
- **CPU benchmarks:** compared Julia, NumPy/Numba and SISAL implementations on Linux using BenchmarkTools.jl and hyperfine, exploring multithreading, vectorisation and compiler settings.
- **Personal Linux & cloud projects:** ran K3s on Debian VMs using QEMU/KVM and deployed an AWS-hosted portfolio using S3, CloudFront, Route 53 and Lambda.